
Evolvability of marine dispersal traits related to larval swimming and reproductive strategy

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Abstract

Analyses of evolvability are often conducted without reference to explicit selection gradients. Additionally, analyses on dispersal evolution focus on selection without reference to multivariate evolvability. Therefore, we compared the evolvability of dispersal traits by: 1) averaging across selection gradients in all directions, 2) considering specific directions of selection hypothesized from biomechanical and life history reasoning, and 3) empirically estimating selection gradients from the genetic covariance between dispersal traits and a component of fitness (settlement). We used an experimental pedigree in the solitary ascidian *Molgula occidentalis* and constructed Bayesian quantitative genetic models to estimate evolvability. There was greater additive genetic variation in tail length than in trunk length, as there was for maternal effect and dominance variance. As a result, selection for longer tails (faster swimming) was associated with a higher evolvability compared to selection for shorter tails and longer trunks (energetic efficiency) or shorter tails and trunks (fecundity). Based on an observed selection gradient, evolvability was still greater for tail length compared to trunk length, even though selection was stronger on trunk length. Overall, evolutionary change in dispersal distances would occur more rapidly under selection on swimming ability than on energetic economy and fecundity. Furthermore, independent lineages of Molgulidae have repeatedly diverged along the larval morphology axis in which we measured the highest evolvability. Our results demonstrate the utility of combining estimates of evolvability with hypothesized or observed selection gradients.

Introduction

Dispersal moves individuals and genes from one place to another, so it is a fundamental determinant of gene flow that mediates evolutionary responses to selection that varies in space and time (Olivieri and Gouyon 1997; Kokko and López-Sepulcre 2006; Garant et al. 2007). Dispersal itself can also be the target of selection, altering patterns of gene flow (Phillips et al. 2006). There is well-established theory for why selection can favor dispersal, despite the costs of movement, that relate to avoiding inbreeding and competition with kin, and to spreading risk when competition and habitat quality varies in both space and time (Johnson and Gaines 1990; Bowler and Benton 2005; Starrfelt and Kokko 2012). Understanding the evolution of dispersal therefore has wide-ranging impacts for understanding population adaptation more generally.

However, simply knowing what selects for dispersal is not enough to predict the capacity for dispersal to evolve (i.e., its evolvability) (Burgess et al. 2016). In natural populations, the distribution of distances that offspring settle from their parents emerges as an outcome of multiple physiological, morphological, life history, and behavioral traits that interact with each other and with wind, currents, movement vectors, competitors, and habitat quality (Clobert et al. 2009; Ronce and Clobert 2012; Stevens et al. 2013). Evolutionary change in the properties of the dispersal distribution will occur only to the extent that there is heritable genetic variation in the direction of selection on the underlying phenotypes that affect the displacement between parents and offspring (Burgess et al. 2016; Saastamoinen et al. 2018). Therefore, a focus on variation and constraint in the many traits that cause dispersal, and not just the emergent dispersal distances, is needed to properly understand the evolvability of dispersal in natural populations. Yet, knowledge of dispersal evolvability remains scarce, and studies on the genetic basis of

dispersal are mostly skewed to single traits in a few groups in terrestrial systems, notably insects, birds, and plants (Saastamoinen et al. 2018).

The evolvability of dispersal can be assessed and compared using standardized metrics that reflect standing genetic (co)variation (the **G** matrix) in the multivariate set of traits that affect dispersal and fitness for a given species (Houle 1992; Hansen and Houle 2008; Hansen 2023). Hansen and Houle (2008) proposed that evolvability in multivariate phenotype space can be measured in numerous ways. *Unconditional evolvability* is the expected magnitude of the evolutionary response in the direction of selection (i.e., towards the fitness peak on an adaptive landscape) when all traits are free to change according to the genetic covariances specified in the **G** matrix. *Conditional evolvability* is the expected magnitude of the evolutionary response in the direction of selection when some traits are constrained from changing, so that only the variation independent of those traits contributes to the response (Hansen et al. 2003). In the context of dispersal evolution, conditional evolvability is particularly relevant because selection can act on traits affecting dispersal for reasons unrelated to dispersal itself, such that traits are not free to respond to selection for dispersal (Burgess et al. 2016). *Responsibility* is the expected magnitude of the evolutionary response in the direction that the population mean phenotype actually moves as it climbs the fitness peak on the adaptive landscape, which is determined by genetic covariances so is not always in the direction of selection (Hansen et al. 2023). These metrics provide a comparable, multivariate view of the capacity for dispersal to evolve, at least in the short-term (microevolution). In the long term (macroevolution), evolvability potentially predicts phenotypic divergence among populations and species (Holstad et al. 2024; Tsuboi et al. 2024). However, most analyses of evolvability are conducted without explicitly incorporating knowledge about the specific direction of selection. An unstructured **G** matrix means that

68 evolvability will not be the same in every direction of selection. A lower dispersal evolvability
69 will therefore indicate constraints that result in a failure to sufficiently adapt to changes in the
70 factors selecting for dispersal (Gomulkiewicz and Houle 2009), and ultimately potential lineage
71 extinction (Jablonski and Lutz 1983; Krug et al. 2015).

72 Understanding the evolvability of dispersal is complicated by the fact that many
73 dispersal-related traits are also under selection from reproductive and developmental processes,
74 and are shaped by maternal effects, which affect the genetic architecture (e.g., pleiotropy,
75 epistasis) underling their covariation. Across many taxa, dispersal occurs early in life during
76 development (e.g., seeds of plants and larvae of benthic marine fishes and invertebrates), or soon
77 after development (e.g., natal dispersal in insects, reptiles, birds, and mammals), when maternal
78 effects and inbreeding depression are expected to be strongest (McNamara et al. 2016; Bell and
79 Hellmann 2019). Maternal effects not only bias estimates of evolutionary potential if
80 unaccounted for (Kruuk and Hadfield 2007), but can themselves be the target of selection
81 (Kirkpatrick and Lande 1989; Räsänen and Kruuk 2007; McGlothlin et al. 2010; Wolf and Wade
82 2016). Furthermore, offspring size and number both affect dispersal outcomes but also depend
83 on the broader problem of how organisms allocate resources across their life history, for example
84 to reproduction vs growth or survival, and then how resources for reproduction are divided
85 between many smaller offspring or fewer larger offspring. Producing more offspring may
86 increase dispersal and better sample different environments, thereby spreading the risk of failure
87 in variable and unpredictable environments (Norghauer and Newbery 2015; Schreiber and
88 Beckman 2020). Smaller (and hence more numerous) offspring may also lead to more efficient
89 dispersal in wind-dispersed seeds (Clark and Ji 1995). On the other hand, investment into
90 morphological structures (such as samaras, fruits, or flight muscles) or energy reserves that

facilitate dispersal may trade off against fecundity (Zera and Denno 1997; McCulloch et al. 2025). Therefore, selection on the traits causing dispersal can occur in many directions of phenotypic space from multiple causes, and evolvability of such traits will not necessarily be the same in all directions.

Here, we sought to understand the evolvability of dispersal-related traits by focusing on traits that are common across ascidians as a model (phylum: Chordata). As adults, ascidians are sessile, but relatively brief periods of dispersal (minutes to hours depending on the species) occur during the larval stage, which is hypothesized to occur for dispersal and habitat selection (Cloney 1982; Svane and Young 1989; Strathmann 2007). Larvae superficially resemble the tadpole larvae of frogs and have a rigid globulose trunk (sometimes called a ‘body’ or ‘head’) that is propelled through the water by a tail. Across species, tail and trunk sizes vary by about an order of magnitude and mean tail size strongly covaries positively with mean trunk size (McHenry and Patek 2004). A reasonable expectation, therefore, is that tail size and trunk size should exhibit positive additive genetic covariation within a population (Schluter 1996; McGlothlin et al. 2018; Tsuboi et al. 2024). However, different causes of selection are expected to increase or decrease tail and trunk simultaneously or change tail size relative to trunk size (McHenry and Patek 2004). Longer tails allow larvae to swim faster (McHenry and Patek 2004), and swimming speed can either increase or decrease dispersal distance. For example, swimming faster would increase dispersal distance in weaker currents when there is a need to move away from predators, kin, competitors, or unfavorable conditions towards better conditions (Bowler and Benton 2005). On the other hand, swimming faster would decrease dispersal under stronger water currents when there is a need to avoid being advected away from suitable habitat (Byers and Pringle 2006; Burgess et al. 2022).

Selection to swim faster (a in Figure 1) would mean that dispersal evolvability would depend more on the genetic variation in tail length compared to that in trunk length. In contrast, relatively shorter tails and larger trunks (b in Figure 1) allow larvae to reduce the energetic costs of swimming at the expense of swimming faster (McHenry and Patek 2004), such that a positive genetic covariance between tail and trunk size would represent a constraint on dispersal evolvability. Larvae that reduce the energetic costs of swimming would have higher energy reserves to delay settlement and disperse further in currents or choose better settlement habitat, and would have higher post-settlement survival (Marshall et al. 2003; Pechenik 2006; Burgess et al. 2012), since ascidian larvae do not feed and rely on a fixed amount of maternally provided resources. Finally, relatively shorter tails and trunks would allow mothers to produce more offspring, because of offspring size-number trade-offs (Smith and Fretwell 1974). Selection for increased fecundity (c in Figure 1) would mean that dispersal evolvability would be highest for a positive genetic covariance between tail and trunk size; opposite to that predicted if selection favored swimming efficiency. Therefore, selection could favor different combinations of trunk and tail size (Huber et al. 2000; McHenry and Patek 2004), shaping both the structure of the **G** matrix and the response to future selection, necessitating a multivariate approach to understand the evolvability of dispersal.

The goal of this study was to characterize the sources of variation in dispersal-related traits to understand: 1) how fast could dispersal-traits evolve, 2) if evolvability would differ under *a-priori* hypotheses about how selection would favor different combinations of trunk and tail size (Figure 1), and 3) evolvability under the observed effect of trunk and tail lengths on settlement probability. We used an experimentally tractable system (the solitary ascidian *Molgula occidentalis*) to conduct a full-sib, half-sib breeding design and constructed Bayesian

quantitative genetic models to partition the phenotypic variance of dispersal-related traits into additive genetic variance, maternal effect variance, and dominance variance. We focused on two morphological traits (the length of the larval trunk and tail) that are expected to affect dispersal distance (McHenry and Patek 2004) and two fitness-related traits (hatching success and settlement success) that affect the number of dispersing offspring at the emigration and immigration stage, respectively, of the dispersal process.

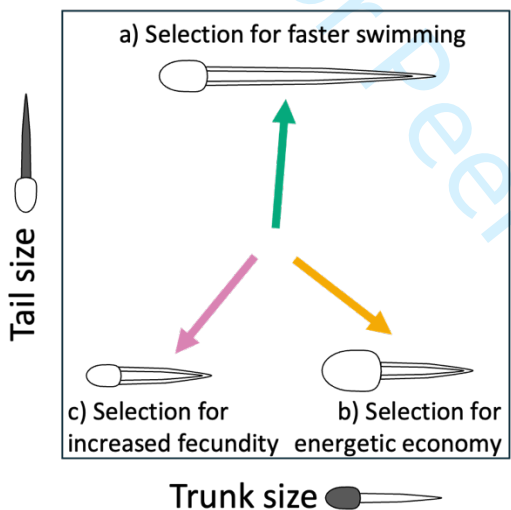


Figure 1: Predicted direction of selection on larval ascidian trunk and tail size when there is a) selection for faster swimming (larger tails; green), b) selection to reduce the energetic costs of swimming (smaller tails, larger heads; yellow), and c) selection for increased fecundity (smaller tails, smaller trunks; purple). Concept and figure adapted from McHenry and Patek (2004).

Methods

Study species

Molgula occidentalis (Traustedt 1883) is a sessile, solitary ascidian native to sand flats and seagrass beds in the Northern Gulf of Mexico, USA. Mature *M. occidentalis* are simultaneously hermaphroditic – individuals can produce both sperm and eggs at the same time (Berrill 1931). External fertilization occurs via the release of sperm and eggs into the water column. The embryos develop in the water column without parental care before hatching into tadpole-like larvae after a few hours. Like many ascidians, the larvae of *M. occidentalis* display photokinetic responses to light (Berrill 1931; Svane and Young 1989), where light arrests movement and removing light induces swimming. Also like many ascidians, *M. occidentalis* larvae exhibit a helical swimming pattern, with pitching and yawing movements around a center of mass located near the intersection of the tail and trunk (McHenry 2005). In other species of ascidians, larvae often swim upwards or directionally towards light for a brief period before swimming downward away from the light to settle, or they simply alternate between periods of active swimming with periods of sinking or inactivity (Olson 1983; Chia et al. 1984; Young 1986, 1995; Svane and Young 1989; Bostwick et al. 2020).

The trunk contains maternally derived energy, organs used for gathering sensory information (e.g., light and gravity), and cell types that contribute to adult organs after metamorphosis that allow the larvae to locate a suitable settlement site (Olson 1983; Jeffery 2004; Tarjuelo and Turon 2004; Kusakabe and Tsuda 2007; Satoh 2014). The tail contains the notochord and muscle fibers that are used to generate thrust for swimming and most of these tissues are resorbed after dispersal at metamorphosis (Swalla and Jeffery 1996; Takada et al. 2002; McHenry et al. 2003; McHenry and Patek 2004; McHenry 2005; Jiang and Smith 2007; Fodor et al. 2021). In non-feeding larvae, the energy used for swimming and metamorphosis is provided by the mother. Larger larvae with increased energy stores may be able to spend more

time searching for a quality site to settle and have higher settlement success and post-settlement fitness than smaller larvae (Marshall and Keough 2003; Thiyagarajan and Qian 2003; Pechenik 2006; Thiyagarajan et al. 2007; Burgess and Marshall 2014). For example, differences in larval size within a species can result in a 30-fold increase in the amount of time spent swimming, and therefore dispersal potential (Marshall and Keough 2003). Furthermore, trunk and tail size vary greatly among ascidian species (especially between solitary vs colonial species), and in some groups, tailless larvae have independently evolved multiple times (Huber et al. 2000; McHenry and Patek 2004).

Collection of adults

M. occidentalis adults were collected from seagrass beds (mostly *Thalassia testudinum* and *Halodule wrightii*) or sandy bottom habitats on the north shore of Dog Island (29°49'42.0"N 84°34'50.9"W) in the Gulf of Mexico, offshore from the Florida State University Coastal and Marine Lab (FSUCML) in Florida USA, between March and October 2019. The collection sites were subtidal, ranging from 0.2 – 1.5 meters in depth. The adults were transported to Florida State University in coolers filled with seawater collected at the site.

Collection of gametes

To conduct the breeding design, gametes were obtained by physically removing them from the gonads of the field collected adults. All gametes were collected within 72 hours from when adults were collected from the field. Though the species is hermaphroditic, it is easy to separate sperm and eggs. Individuals were randomly chosen to provide eggs or sperm, and eggs

were collected from different individuals than sperm were collected. Individuals that provided sperm were called sires and individuals that provided eggs were called dams.

Eggs from dams were collected by removing the tunic with medical scissors, gently cutting the ovaries with a scalpel, and flushing the incision with filtered seawater (autoclaved, 8 μm filter, and ranging from 28 - 31 ‰ in salinity) to remove eggs from the ovary. Eggs were gently rinsed with filtered seawater using a 63 μm mesh filter (eggs were typically 138-159 μm in diameter) to remove debris. After the eggs were rinsed, they were placed in a 60 mm petri dish with 4 ml of filtered seawater to create egg stocks from each dam.

Sperm from sires were collected by removing the gonads (both ovaries and testes) from the body wall and mincing the gonads using scissors and forceps. The minced gonads were rinsed through a 30 μm mesh filter with filtered seawater so that eggs and other fragments of the gonads were caught on the filter while water containing sperm passed through the mesh into a 60 mm petri dish. To avoid cross contamination, different sets of tools were used for each individual. Sperm stock solutions of each sire were kept separate. Furthermore, each dam had a control dish of eggs to which sperm were not added. Thirty minutes after the eggs were collected, control dishes were inspected under a dissecting microscope and the entire mating block was discarded if a single egg was fertilized.

With each sire $\times$ dam cross, three replicate petri dishes each contained 1 ml of egg stock solution from each mother and 8 ml of filtered seawater. Sperm stock from each sire was added to each mother's petri dishes so that there were 4.0×10^5 sperm cells ml^{-1} in each dish (previous analysis determined that fertilization success was greatest at this concentration; *unpublished data*). After 30 minutes of exposure to sperm, the eggs were washed of sperm by gently rinsing them with filtered seawater over a 63 μm filter to reduce the chances of polyspermy and protect

developing embryos from fouling due to dying sperm cells. After rinsing, the eggs were moved into clean petri dishes filled with 9 ml of filtered seawater.

Breeding design

To estimate quantitative genetic parameters, sperm and eggs from adults were crossed in accordance with a paternal and maternal full-sib and half-sib design with factorial mating (often called a North Carolina II [NCII] breeding design), which allows the independent estimation of additive, maternal, and non-additive sources of variance (Lynch and Walsh 1998). Crosses were performed in temporal blocks over a six-month period. Each temporal block used different sires and dams that were collected from the field up to three days prior to dissection and fertilization of gametes. Within each temporal block, sperm from 2 to 3 sires were crossed with eggs from 2 to 3 dams, resulting in 4 to 9 full-sib families (Table 1). An individual adult served only as a sire or dam. In this breeding design, the l th offspring is the result from a mating between the i th sire (s_i) and the j th dam (d_j). All full-sib offspring of sire s_i and dam d_j compose the interaction I_{ij} , and all of the interactions of those sires and dams occur in the k th temporal block. For a single trait, the phenotype z of an offspring from the i th sire and j th dam from the k th block is

$$z_{kij} = \mu_k + s_i + d_j + I_{ij} + e_{ijl}, \quad (1)$$

where μ_k is the mean phenotype in that offspring's temporal block. s_i is the effect of i th sire, d_j is the effect of the j th dam, I_{ij} is the effect due to the non-additive effects of parents i and j , and e_{ijl} is the deviation of the offspring's phenotype from that predicted from parents i and j .

Variance components can be expressed in terms of hypothesized underlying causal factors (Lynch and Walsh 1998). The variance in sire effects s_i is $\frac{1}{4}$ of the variance associated with the average additive effect of alleles (additive genetic variance, V_A). The dam effect

includes the additive effects of alleles inherited from the mother as well as genetic or environmental maternal effects, so it will be higher than the sire effect when there are maternal effects. Therefore, the variance in $d_j - s_i$ provides an estimate of the maternal effects variance V_M (i.e., effects on offspring phenotype due to the maternal environment or genotype beyond the additive effect of inherited maternal alleles). The variance in the interactive effects I_{ij} of paternal and maternal alleles is associated with the non-additive effect of alleles and is $\frac{1}{4}$ of the variance due to dominance effects (V_D) (Lynch and Walsh 1998).

Table 1: Summary of the number of samples used in the models for each trait from each component of the breeding design.

Trait	No. of blocks	No. of sires	No. of dams	No. of full-sib families	Mean ($\pm$ sd) no. of measurements per full-sib family	Total no. of individuals measured
Trunk and tail length	15	39	40	106	10.9 ($\pm$ 2.67)	1,152
Hatching	10	28	29	81	47.5 ($\pm$ 10.7)	3,846
Settlement	10	26	27	71	54 ($\pm$ 9.02)	3,836

Measurement of phenotypes

Hatching success was measured as whether an embryo developed into a larva 12 – 14 hours after fertilization. Immediately after fertilization, approximately 20 embryos (confirmed visually under a dissecting microscope by the presence of cleavage or multiple cells within a single egg) from each of three dishes per full-sib family were relocated to new dishes. After 12 – 14 hours, individuals were scored as 1 if a larva and 0 if still an embryo.

Settlement success was measured as whether a larva attached to the petri dish 25 to 30 hours after hatching success was measured. Settlement was scored on different larvae within

each cross to which hatching was scored. After hatching, approximately 20 larvae from each of the three dishes per full-sib family were each placed into their own dish and settlement success was measured 25 to 30 hours later.

Trunk length was quantified as the distance between the front of the trunk and the intersection of the tail and trunk. Tail length was quantified as the distance along the notochord and muscle tissue. Trunk and tail length (from the same larvae) were scored on different larvae within each cross to which hatching and settlement success were scored. Larvae were photographed with a scale under a dissecting microscope fitted with a ZEISS Zen camera 1 to 4 hours after hatching success was measured (13 to 18 hours after fertilization). Larvae cease swimming under bright light, which facilitated image capture while larvae were stationary on the bottom of the dish. Images were exported into ImagePro Insight 8.0, where trunk and tail length were measured to the nearest 0.01 μm .

Data Analysis

To partition phenotypic variance and estimate evolvability, we constructed hierarchical Bayesian models to estimate additive (**G**), maternal effect (**M**), dominance (**D**), and residual (**R**) variance-covariance matrices for trunk and tail lengths (Model I: Trunk and Tail), the ratio of tail to trunk length (Model II: Trunk:Tail Ratio), and hatching and settling of larvae (Model III: Hatching and Settling). Additionally, we created a fourth model to predict the evolutionary response for trunk or tail length when the probability of settling determines fitness (Model IV: Genetic covariance between traits and relative fitness). The advantage of a Bayesian approach is that estimates of variances and their uncertainty properly account for the skewness that arises because variances cannot be less than 0, as well as the propagation of uncertainty to derived

estimates like evolvability, which are not accommodated in frequentist approaches (de Villemereuil 2019).

All four models were similar in construction, and the same terminology is used throughout. Below, we start by detailing Model I, which estimates the above variance-covariances matrices for trunk and tail lengths, followed by a description of modifications for the other three models. All models were run in R v4.1.2 (R Core Team 2022) using the *runjags* v2.2.2-5 package (Denwood 2016). For all models, 50 MCMC chains were run, with 100 adaptation steps. The burn-in length, total number of posterior samples, and thinning rate were set for each model independently to achieve good mixing, convergence, and effective sample sizes ≥ 5000 for each parameter of interest. MCMC chains were checked for convergence and mixing by examining their traces by eye, autocorrelation was around 0, and the Gelman-Rubin potential scale reduction factor for all parameters of interest was $\hat{R} < 1.02$. We also conducted posterior predictive checks for all parameters of interest which showed that model predictions were consistent with the data. Summaries of the MCMC diagnostics and posterior predictive distributions are provided in the Supplemental Materials 1. A comparison of **G**, **M**, **D**, and **P** using the same model structure (Model I) implemented in WOMBAT (Restricted Maximum Likelihood) and MCMCglmm (Bayesian), as well as a model in MCMCglmm using blocks as random factors, can be found in Supplemental Materials 2.

Our goal was to quantify the posterior probability distribution of the variance components and evolvability, rather than to describe variances and evolvabilities with respect to a null hypothesis that they are zero. Posterior distributions were summarized using the mode for the central tendency (using the Venter method implemented using the *mlv()* function from the package ‘modeest’ v2.4.0, (Poncet 2019)), the 95% highest posterior density interval, HPDI, for

the uncertainty (implemented using the HDI() function from the package ‘HDInterval’ v0.2.4 (Meredith and Kruschke 2022)), and the probability that covariances were positive (greater than 0). The 95% HPDI is the interval which contains 95% of the distribution mass, such that all points within the interval have a higher probability density than points outside the interval (Kruschke 2011). R code and data for all models and figures are available from: *[the anonymized repository is available on the submission site as Bayesian-evolvability-R1.zip]*.

Model I: Trunk and Tail

Based on the linear predictor in equation (1), the likelihood of observing the combined trunk length for the l th offspring (Θ_l) was drawn from a Multivariate Normal distribution (MVN):

$$\Theta_l \sim MVN(\mu_k + s_i + d_j + I_{ij}, e_{ij})$$

where for trait = 1 (trunk) or 2 (tail):

$$s_i \sim MVN\left(0, \begin{bmatrix} \sigma_{s1}^2 & \sigma_{s12} \\ \sigma_{s12} & \sigma_{s2}^2 \end{bmatrix}\right) = V_{sire}$$

$$d_j \sim MVN\left(0, \begin{bmatrix} \sigma_{d1}^2 & \sigma_{d12} \\ \sigma_{d12} & \sigma_{d2}^2 \end{bmatrix}\right) = V_{dam}$$

$$I_{ij} = MVN\left(0, \begin{bmatrix} \sigma_{I1}^2 & \sigma_{I12} \\ \sigma_{I12} & \sigma_{I2}^2 \end{bmatrix}\right) = V_{interaction}$$

$$e_{ij} = MVN\left(0, \begin{bmatrix} \sigma_{e1}^2 & \sigma_{e12} \\ \sigma_{e12} & \sigma_{e2}^2 \end{bmatrix}\right) = V_{residual}$$

and μ_k is a matrix of block phenotypic means (n blocks $\times$ n traits). From the posterior sample of V_{sire} , V_{dam} , $V_{interaction}$, and $V_{residual}$, we computed the following (Lynch and Walsh 1998):

$$\text{Additive genetic variance-covariance, } \mathbf{G} = 4 \times V_{sire}$$

$$\text{Maternal effect variance-covariance, } \mathbf{M} = V_{dam} - V_{sire}$$

Dominance genetic variance-covariance, $\mathbf{D} = 4 \times V_{interaction}$

Residual variance-covariance, $\mathbf{R} = V_{residual}$

Total phenotypic variance, $\mathbf{P} = \mathbf{G} + \mathbf{M} + \mathbf{D} + \mathbf{R}$.

Priors

Priors for the mean value of the trunk and tail length phenotypes within each block (μ_k) were drawn from Uniform distributions spanning the minimum and maximum range of each of the two respective phenotypes in the dataset. Hyperpriors for the sire effects, dam effects, and non-additive effects for each phenotype were two-dimensional vectors drawn from Normal distributions with a mean of zero and variance of 10^5 . Priors for the variances (diagonal elements: σ^2) were Uniform distributions between 0 and 1000. The covariances (off-diagonal elements) were computed by first estimating a value for the correlation coefficient between the effects ($r_{x,y}$), with a prior distribution defined as $r_{x,y} \sim Uniform(-1, 1)$. The covariance between effects for the two phenotypes ($cov_{x,y}$) was then computed as $cov_{x,y} = r_{x,y} \cdot \sqrt{var_x \cdot var_y}$.

Model II: trunk:tail Ratio

In model II, we focused on the ratio of trunk to tail length in each offspring, as this measure captures relative size, or size allometry. Given that this ratio is log-normally distributed, the model used the log-transform of the ratio and parameters are estimated in log-space. We estimated similar parameters as in Model I, but because there was only a single phenotype, we only estimated the additive effect, maternal effect, non-additive, and residual variance rather than

variance-covariance matrices. Priors on these variances and hyperpriors on block means, and additive, maternal, and non-additive effects were distributed as described in Model I.

Model III: Hatching and Settling

In Model III, we focused on two fitness-related traits – hatching success affects the number of dispersing offspring at the beginning of dispersal, while settlement success affects the number of dispersing offspring that complete the dispersal phase. Model III was designed to estimate the same variance-covariance matrices as in Model I, but the responses being modeled were the hatching and settling of offspring. Hatching and settling were both scored on individual offspring as 0 for failure and 1 for success (ρ_θ), and were evaluated for different offspring of each of the three replicates of the same family. Settlement failure or success was assessed on offspring that hatched. The likelihood of hatching or settling was computed from Bernoulli distributions given an estimated probability of hatching or settling,

$$\rho_\theta \sim \text{Bernoulli}(\text{Pr}(\theta)).$$

The log odds ratio (logit) of the estimated probability of hatching or settling for an offspring of the i -th sire, and j -th dam was defined as the sum of the block mean, sire and dam additive effects, maternal effects, and non-additive effects as in Model I,

$$\text{logit}(\text{Pr}(\theta)) = \mu_b + s_i + d_j + I_{ij}.$$

The total phenotypic variance was then $\mathbf{P} = \mathbf{G} + \mathbf{M} + \mathbf{D}$, since the residual variance is determined by the mean in a Bernoulli model. Priors for means and variance-covariance matrices on the logit scale were defined as in Model I.

Converting parameters estimated on the latent scale to the data scale

In Model II and III, the traits are non-Gaussian traits and are not additive on the scale on which they are expressed and selected (de Villemereuil et al. 2016). Variances estimated on the latent scale (log ratio in Model II and logit in Model III) were converted to variances on the expected data scales (ratio in Model II and probability in Model III) following the framework developed by de Villemereuil et al. (2016) and implemented using the package ‘QGglmm’ v0.8.0 (de Villemereuil et al. 2016). The package takes estimates of variances and covariances on the latent scale, as well as the predicted block means on the latent scale, and returns estimates on the observed scale. We used the function QGparams() for Model II and QGmvparams() for Model III.

Model IV: Genetic covariance between traits and relative fitness

In this model, we only used the 10 blocks in which trunk length, tail length, and settlement were measured for the same families (i.e., 10 blocks, Table 1). Following Stinchcombe et al. (2014), we combined the approaches in Model I and III to estimate a 3x3 matrix, denoted $\mathbf{G}_{zw}$, where the rows and columns are trunk length i , tail length j , and the probability of settlement w :

$$\mathbf{G}_{zw} = \begin{bmatrix} \sigma_i^2 & \sigma_{ij} & \sigma_{iw} \\ \sigma_{ji} & \sigma_j^2 & \sigma_{jw} \\ \sigma_{wi} & \sigma_{wj} & \sigma_w^2 \end{bmatrix}.$$

Variance and covariances for the probability of settlement were estimated on the logit scale but converted to the probability scale using QGmvparams() accounting for block effects. From $\mathbf{G}_{zw}$, the additive genetic covariance between trunk length and the probability of settlement (e.g., σ_{iw}), and between tail length and the probability of settlement (e.g., σ_{jw}) indicate the genetic selection differentials (Stinchcombe et al. 2014). To get the selection differentials in

units of phenotypic change per generation based on the Robertson–Price Identity (Lynch and Walsh 1998; Stinchcombe et al. 2014), where relative fitness is $w = p/\bar{p}$, we just divide the covariance by the mean of p , $\bar{p}$:

$$\mathbf{s}_g = \begin{bmatrix} \text{cov}(w, z_{trunk}) = \sigma_{iw}/\bar{p} \\ \text{cov}(w, z_{tail}) = \sigma_{jw}/\bar{p} \end{bmatrix}.$$

If we consider $\mathbf{G}$ to be the upper 2x2 of $\mathbf{G}_{zw}$ (containing the variances and covariances for trunk and tail only), the selection gradient is then $\boldsymbol{\beta} = \mathbf{G}^{-1}\mathbf{s}_g$.

Evolvability

Following Hansen and Houle (2008) and Hansen (2023), the additive genetic variance matrix $\mathbf{G}$ is mean standardized, and the vector of directional selection gradients (in this case, for two traits) is standardized to unit length $\boldsymbol{\beta} = \sqrt{\boldsymbol{\beta}_1^2 + \boldsymbol{\beta}_2^2} = \mathbf{1}$. When $\boldsymbol{\beta} = \mathbf{1}$, a proportional change in the phenotype yields the same proportional change in relative fitness. Unconditional evolvability, $e(\boldsymbol{\beta}) = \boldsymbol{\beta}^T \mathbf{G} \boldsymbol{\beta}$, gives the predicted proportional change in the trait mean per generation in the direction of the selection gradient, under unit selection. Conditional evolvability, $c(\boldsymbol{\beta}) = (\boldsymbol{\beta}^T \mathbf{G}^{-1} \boldsymbol{\beta})^{-1}$, gives the predicted proportional change in the trait mean per generation in the direction of the selection gradient if the response was constrained to follow the gradient by selection against deviations. Responsability, $r(\boldsymbol{\beta}) = \sqrt{\boldsymbol{\beta}^T \mathbf{G}^2 \boldsymbol{\beta}}$, gives the predicted proportional change in the trait mean per generation in the direction of the selection differential. The ratio between the conditional evolvability and the unconditional evolvability $c(\boldsymbol{\beta}) / e(\boldsymbol{\beta})$ measures the fraction of genetic variance in the direction of $\boldsymbol{\beta}$ that is free from constraints.

We estimated evolvability parameters using three approaches. In the first approach, we used the **G** matrix from Model I and Model III on the expected data scale with randomly generated selection gradients. In the second approach, we used the **G** matrix from Model I with the expected (fixed) selection gradients based on the *a-priori* hypotheses in Figure 1. In the third approach, we computed the percent change per generation directly from the mean standardized genetic selection differential from Model IV.

In the first approach, we assumed no knowledge about the direction of selection that the population experiences, so we assessed the average evolutionary potential of the **G** matrix over random directional selection gradients, β (a column vector of partial regression coefficients standardized to unit length). We used the R package called ‘evolvability’ v2.0.1 to generate the selection gradients (Bolstad et al. 2014). Posterior distributions of evolvability parameters from a set of selection gradients were calculated using the function `evolvabilityBetaMCMC()`. We estimated evolvability parameters over 1,000 randomly generated selection gradients in a 2-dimensional space using the function `randomBeta()`, which produces unit-standardized selection gradients that are equally likely in all directions (uniform distribution on the surface of a sphere) by utilizing the spherically symmetrical properties of the Gaussian distribution. Evolvability estimates for probability traits (Model III) cannot be extrapolated linearly over multiple generations due to bounding between 0 and 1 (Morris and Doak 2004; Pélabon et al. 2020), therefore comparisons with Model I are restricted to one generation and contingent on the current mean probability of hatching and settling.

In the second approach, we considered three scenarios based on a-prior expectations outlined in Figure 1, while maintaining unit length standardization:

1) selection for longer tails only (faster swimming) where $\beta = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$,

- 2) selection for shorter trunks and shorter tails (increased fecundity) where $\beta =$
- $$\begin{bmatrix} -1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix},$$
- 3) selection for shorter trunks and longer tails (energetic economy) where $\beta = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}.$

Posterior distributions of evolvability parameters from each set of selection gradients were calculated using the function `evolvabilityBeta()`. This approach accounts for the fact that evolvabilities will not be the same in different directions of phenotype space when there are genetic covariances.

In the third approach, we used empirically estimated genetic selection differentials from Model IV to directly compute the expected evolutionary response in the direction of selection, which can be expressed for each trait $i = 1,2$ as

$$\frac{\Delta \bar{z}_i}{\bar{z}_i} = \frac{s_{g,i}}{\bar{z}_i}$$

and when multiplied by 100 gives the percent change in the population trait mean per generation.

Results

Model I: Trunk and Tail

Larvae with longer tails tended to have longer trunks; the phenotypic covariance of trunk and tail length was $91.62 \mu\text{m}^2$ ($50.34 - 150.10$, 95% HPDI) (Figure 2, Table 2). The phenotypic mean and variance of tail length was greater than that for trunk length; the mean tail length was $451.15 \mu\text{m}$ ($449.79 - 452.61$, 95% HPDI) with a phenotypic variance of $657.30 \mu\text{m}^2$ ($525.30 - 857.80$, 95% HPDI), and the mean trunk length was $204.58 \mu\text{m}$ ($203.87 - 205.22$, 95% HPDI) with a phenotypic variance of $134.1 \mu\text{m}^2$ ($111.50 - 166.30$, 95% HPDI). Despite this, the mean

standardized additive genetic variance for tail length was still 2.23 (0.24 – 39.45, 95% HPDI) times greater than that for trunk length.

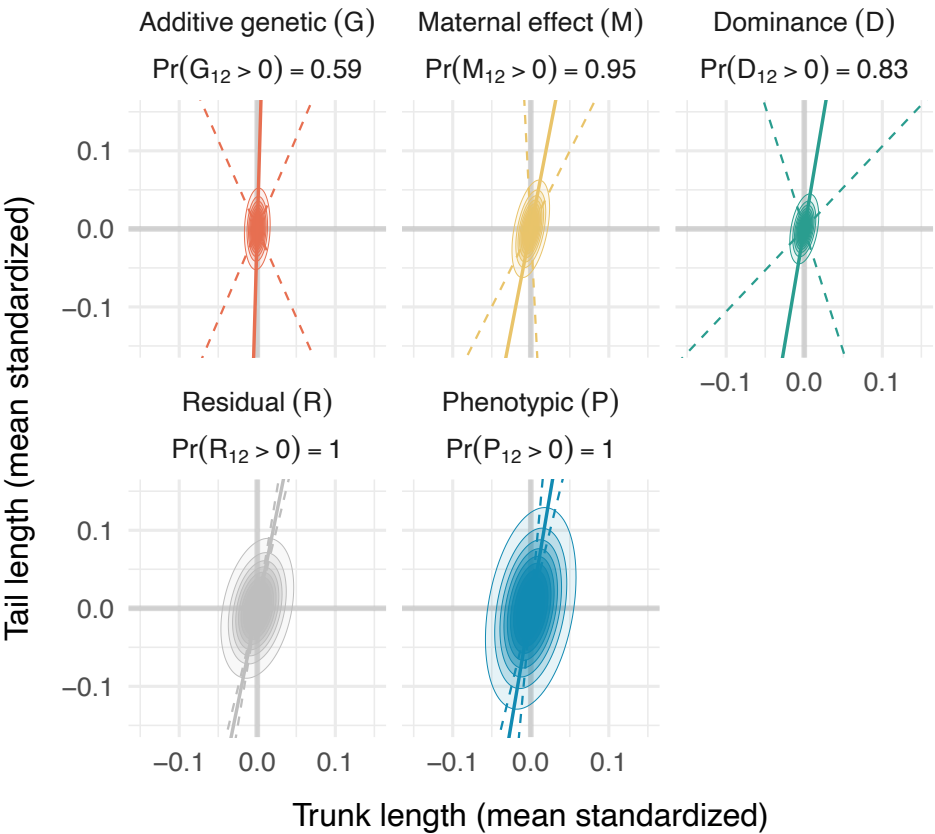
The direction of the additive genetic covariance between trunk and tail lengths was uncertain (Table 2): the posterior mode was close to zero (1.18, -25.7 – 36.73, 95% HPDI) and there was only a 59% posterior probability that the additive genetic covariance was positive. The 95% HPDI for the dominant eigenvector of **P** fell within that for **G** (Figure 2), which means we cannot rule out the possibility that there was not enough information to estimate the small positive covariance in **G** (from 39 sires) that would be predicted from **P** (from 1,152 larvae).

For the maternal effect covariance and the dominance covariance, there was a 95% and 82% posterior probability, respectively, that the covariance was positive, although the 95% highest posterior density intervals included covariances slightly below zero (Table 2). Like that for **G**, the 95% HPDI for the dominant eigenvector of **P** fell within that for **M** and **D** (Figure 2).

For trunk length, additive genetic variance explained only of 3.27% (0.002 – 19.7, 95% HPDI) of the phenotypic variance (Figure 3a), with a 95% probability that heritability was less than 0.2. The variance due to maternal effects was 1.51 times greater than that for additive genetic, with an 83.2% probability that the maternal effect variance was greater than the additive genetic variance.

For tail length, additive genetic, maternal effects, dominance sources of variance explained a slightly higher proportion of the phenotypic variance than these sources of variance did for trunk length (Figure 3b). The additive genetic variance for tail length explained 13.46% (1.94 – 32.4, 95% HPDI) of the phenotypic variance (Figure 3a), with a 95% probability that heritability was less than 0.32. Maternal effect variance explained 22.5% (8.14 – 39, 95% HPDI) of the phenotypic variance in tail length.

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Figure 2: Visual representation of the mean-standardized additive genetic (**G**), maternal effect

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(**M**), dominance (**D**), residual (**R**), and phenotypic (**P**) matrices estimated in Model I. The

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outermost ellipse is the 95% confidence interval of 100 random draws of each posterior sample

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covariance matrix (i.e., 60,000 points). The inner ellipses are in steps of 0.1 from the outer edge

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of 0.95 down to 0.05 in the center. The solid line is the median of the posterior of the major

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eigenvector, and the dashed lines are its 95% highest posterior density interval (HPDI).

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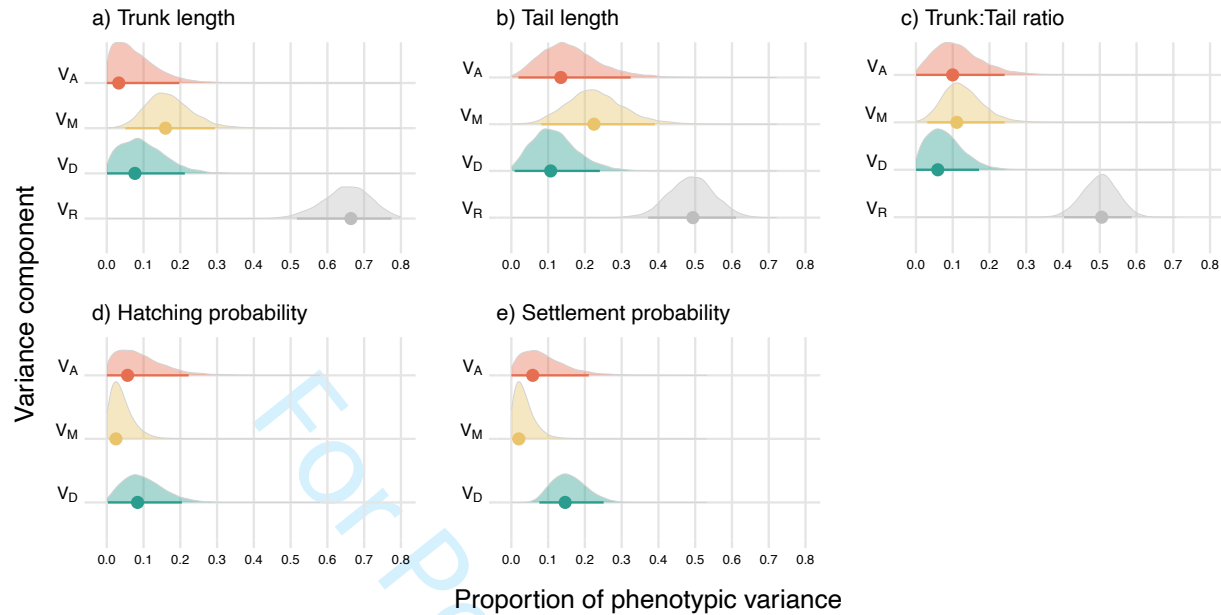


Figure 3: Bayesian posterior distributions for the proportion of phenotypic variance explained by additive genetic variance (diagonals of **G**, denoted V_A), maternal effect variance (diagonals of **M**, denoted V_M), dominance variance (diagonals of **D**, denoted V_D), and residual variance (diagonals of **R**, denoted V_R) for a) trunk length, b) tail length, c) trunk:tail ratio, d) hatching probability, and e) settlement probability. Dots indicate the mode of the distribution. The horizontal bar indicates the 95% highest posterior density interval (HPDI).

Table 2: Estimates of additive genetic variance (**G**), maternal effect variance (**M**), dominance variance (**D**), residual variance (**R**), and total phenotypic variance (**P**) for trunk and tail length (Model I), trunk:tail ratio (Model II), and hatching and settlement probability (Model III). All estimates are on the data scale. The mode and the 95% highest posterior density interval (HPDI) of the posterior distribution is presented. The posterior probability that covariances were positive is also presented.

	G	M	D	R	P
Model I					

Trunk	4.44 (0.00 – 28.70)	19.64 (6.13 – 45.13)	8.96 (0.02 – 31.83)	88.99 (81.77 – 96.99)	134.1 (111.50 – 166.30)
Tail	87.87 (4.29 – 232.9)	127.50 (48.23 – 307.9)	67.71 (1.88 – 167.90)	326.8 (302.4 – 357.6)	657.30 (525.30 – 857.80)
Covariance	1.18 (-25.7 – 36.73) Pr($G_{12}>0$) = 0.59	22.06 (-6.94 – 69.32) Pr($M_{12}>0$) = 0.95	7.51 (-12.24 – 41.83) Pr($D_{12}>0$) = 0.83	49.94 (40.43 – 61.78) Pr($R_{12}>0$) = 1	91.62 (50.34 – 150.10) Pr($P_{12}>0$) = 1
Model II					
Trunk:tail	0.0001078 (0.0000005 – 0.0003146)	0.0001267 (0.0000326 – 0.0003091)	0.0000720 (0.0000000 – 0.0002159)	0.001174 (0.00099 – 0.00143)	0.001174 (0.00099 – 0.00143)
Model III					
Hatching	0.007269 (0.000001 – 0.03855)	0.004036 (0.005856 – 0.01845)	0.01234 (0.000002435 – 0.03444)		0.1581 (0.1369 – 0.1856)
Settlement	0.01419 (0.000014 – 0.05148)	0.004933 (0 – 0.02256)	0.03573 (0.01906 – 0.06191)		0.244 (0.2413 – 0.2462)
Covariance	0.001474 (-0.01478 – 0.0222) Pr($G_{12}>0$) = 0.61	-0.004639 (-0.01539 – 0.002425) Pr($M_{12}>0$) = 0.07	0.006467 (-0.008168 – 0.02109) Pr($D_{12}>0$) = 0.83		0.0039 (-0.0148 – 0.0248) Pr($P_{12}>0$) = 0.65

Model II: Trunk:Tail Ratio

Additive genetic variance for the ratio of trunk to tail length explained 10% (0.07 – 24.1, 95% HPDI) of the phenotypic variance (Figure 3c), with a 95% probability that heritability was less than 0.24. The proportion of phenotypic variance explained by maternal effects and dominance variance was similar to that for additive genetic effects. Trunk lengths were on average 45.6% (42.4 – 47.4, 95% HPDI) of the length of tails.

Model III: Hatching and Settling

On average, 90% (65 – 100, 95% CI) of eggs hatched per full-sib family, and 63% (28 – 95, 95% CI) of larvae settled per full-sib family. There was weak evidence for any phenotypic covariation between hatching and settling, with the probability of the covariance being positive at 65% (Table 2). That is, full-sib families with a higher proportion of eggs hatching did not necessarily have a higher proportion of larvae settling.

There was also weak evidence for additive genetic covariance between hatching probability and settlement probability ($\Pr(G_{12} > 0) = 0.61$; Table 2). There was, however, a 93% probability that the maternal effect covariance was negative (Table 2).

Additive genetic variance in hatching and settlement probability only explained 5.55% (0 – 22.28, 95% HPDI) and 5.81% (0.002 – 21.05, 95% HPDI), respectively, of the phenotypic variance (Figure 3d, e). There was a 95% probability that heritability of hatching and settlement success was less than 0.22 and 0.21, respectively. Dominance genetic variance explained 14.64% (7.63 – 25.1, 95% HPDI) of the phenotypic variance in settlement success.

Evolvability

In terms of the average evolvability of trunk and tail length in the absence of knowledge about the direction of selection, evolvabilities were generally low relative to the median evolvability for traits with linear measurements of about 0.1% estimated by Hansen et al. (2011). The average unconditional evolvability $e(\beta)$ of the trunk and tail $\mathbf{G}$ matrix (i.e., over 1000 unit standardized random selection gradients) was 0.0203 % (0.004 - 0.052, 95% HPDI) (Figure 4). In other words, trunk and tail length would only change by ~0.02% per generation in the

direction of selection, under the strongest selection possible. There was a similar result for
responsability $r(\beta)$, such that trunk and tail length would also change by ~0.02% per generation
in the direction that population mean phenotype moves (Figure 4a).

The conditional evolvability $c(\beta)$ was 0.0098% (0.0005 - 0.0236, 95% HPDI), such that
only 49% (6 – 89, 95% HPDI) of genetic variance in the direction of selection was free from
constraints (i.e., $c(\beta) / e(\beta)$). Conditional evolvability was much lower than unconditional
evolvability since the mean-standardized additive genetic variance was greater for tail length
than that for trunk length.

The average evolutionary potential of the hatching and settling **G** matrix was
approximately 1.75 orders of magnitude greater in the next generation compared to the trunk and
tail **G** matrix (Figure 4). The evolvability of hatching and settling was similar to the 1%
estimated for life history traits by Hansen et al. (2011). The average unconditional evolvability of
the hatching and settling **G** matrix meant that hatching and settling probability would change by
1.14% (0.25 – 3.31, 95% HPDI) of its current value in the next generation. The estimate of
conditional evolvability (Figure 4b) meant that 88% (25 – 100, 95% HPDI) of genetic variance
in the direction of selection was free from constraints.

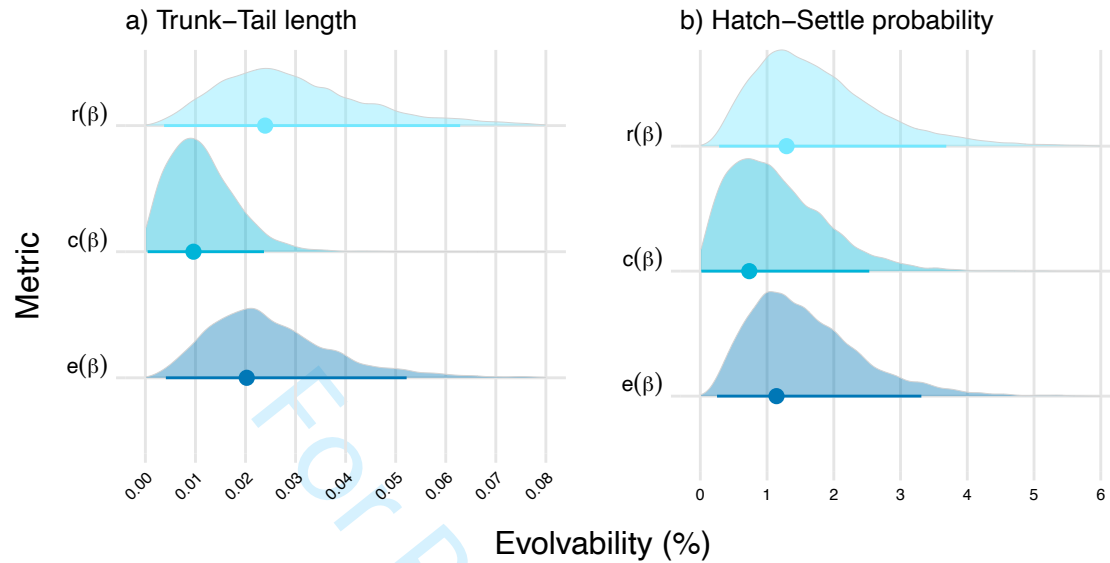


Figure 4: Bayesian posterior distributions for three metrics of evolvability: unconditional evolvability $e(\beta)$, responsibility $r(\beta)$, and conditional evolvability $c(\beta)$ for a) the trunk-tail $\mathbf{G}$ matrix, and b) the hatch-settle $\mathbf{G}$ matrix. Dots indicate the mode of the distribution. The horizontal bar indicates the 95% highest posterior density interval (HPDI). Proportions have been multiplied by 100 to give percent change in phenotype per generation under unit strength of selection.

In terms of evolvability given *a-priori* hypotheses about selection (Figure 1), selection for longer tails (faster swimming) was associated with a higher evolvability compared to selection for shorter tails and longer trunks (energetic economy) or shorter tails and trunks (fecundity), since there was greater mean-standardized genetic variation in tail length than in trunk length (Figure 2). Selection for longer tails (faster swimming) was associated with an unconditional and conditional evolvability of 0.0431% (0.002 - 0.1141, 95% HPDI) and 0.0355% (0.0001 - 0.095, 95% HPDI), respectively (Figure 5). Selection for shorter tails and trunks was associated with conditional evolvability of 0.0102% (0 - 0.0621, 95% HPDI). The

percent of genetic variance free from constraints if selection favored either faster swimming, increased fecundity, or energetic economy was 93.88% (27.02 – 100, 95% HPDI), 87.81% (7.655 – 100, 95% HPDI), and 87.81% (7.65 – 100, 95% HPDI), respectively.

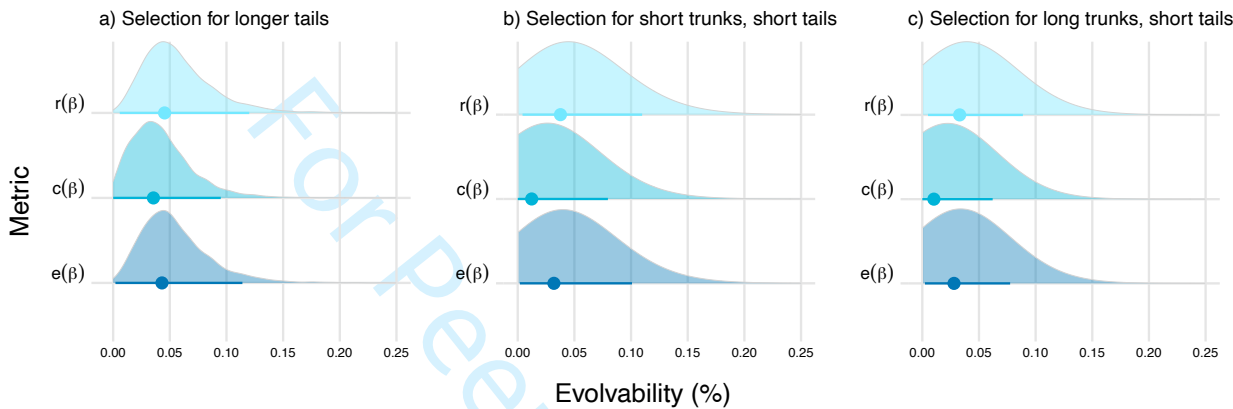


Figure 5: Bayesian posterior distributions for three metrics of evolvability: unconditional evolvability $e(\beta)$, responsibility $r(\beta)$, and conditional evolvability $c(\beta)$ when there is a) selection for longer tails (faster swimming) only, b) selection for shorter trunks and shorter tails (increased fecundity), and c) selection for longer trunks and shorter tails (energetic economy). Dots indicate the mode of the distribution. The horizontal bar indicates the 95% highest posterior density interval (HPDI).

In terms of evolvability given the observed effect of trunk and tail size on the probability of settlement (Figure 6), there was a 72% posterior probability that the expected percent change per generation was positive for both trunk and tail length. The percent change in tail length per generation with the highest posterior probability would be greater than that for trunk length, with trunk lengths increasing by 0.03% (-0.13 – 0.37, 95% HPDI) and tail lengths increasing 0.05% (-0.24 – 0.64, 95% HPDI) per generation. Interestingly, a greater evolvability of tail length

compared to trunk length was expected even though the selection gradient β was relatively steeper for trunks (0.0089, -0.0581 – 0.072, 95% HPDI) compared to tails (0.0028, -0.0099 – 0.0123, 95% HPDI).

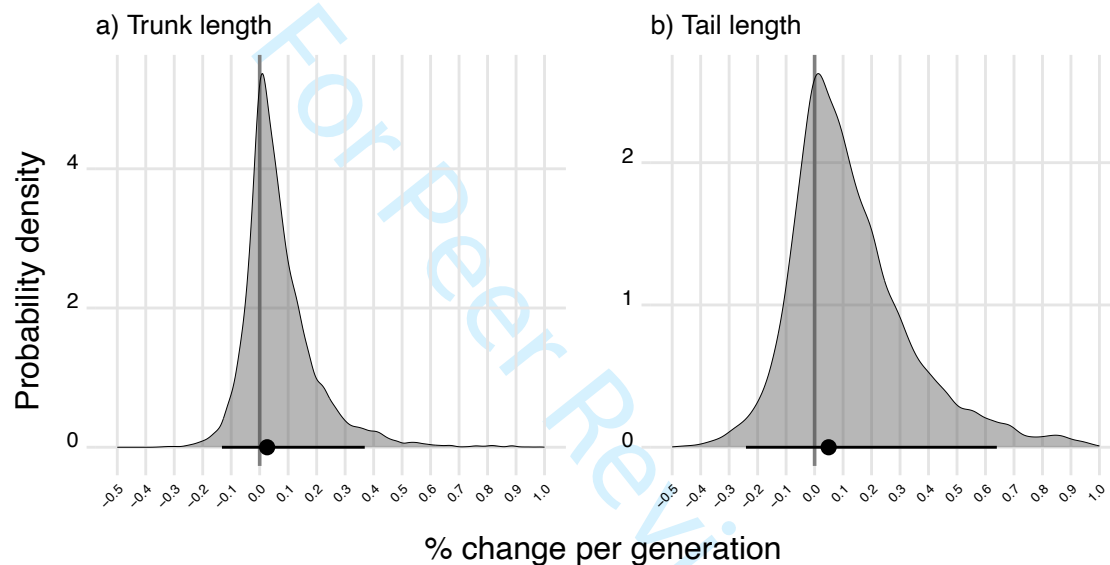


Figure 6: Bayesian posterior distributions for the percent change in the mean a) trunk length and b) tail length per generation if settlement determines fitness, based on the additive genetic covariance between traits and the observed settlement probabilities (Model IV). Dots indicate the mode of the distribution. The horizontal bar indicates the 95% highest posterior density interval (HPDI).

Discussion

There are well-known reasons why selection would favor dispersal (Johnson and Gaines 1990; Bowler and Benton 2005; Starrfelt and Kokko 2012), but much less known on the capacity for dispersal to evolve. Additionally, while there is a robust framework for quantifying

605 evolvability more generally (Hansen et al. 2023), most analyses focus on the pattern of genetic
 606 variation in the **G** matrix without explicit knowledge of the direction of selection gradients
 607 (though see Tsuboi et al. 2024 and references therein). Here, we identified a set of phenotypic
 608 traits that were hypothesized to affect dispersal in ascidians in potentially contrasting ways and
 609 assessed their evolutionary potential using hypothesized and observed selection gradients. Tail
 610 length exhibited a greater evolvability compared to trunk length, even though we estimated
 611 greater (positive) selection on trunk length. Furthermore, there was on average a smaller fraction
 612 of genetic variance in the direction of selection free from constraints in the trunk-tail **G** matrix
 613 (49%) compared to the hatching-settlement **G** matrix (88%). Longer tails allow ascidian larvae to
 614 swim faster (McHenry and Patek 2004; McHenry 2005), and swimming faster could increase
 615 dispersal distance in weaker currents or decrease dispersal distance to avoid advection away
 616 from preferred habitat (Olson 1985; Young 1986; Stoner 1990, 1992). The lower evolvability of
 617 trunk length compared to tail length could also reflect stronger selection in the past on trunk size.

618 Our estimates of evolvability for trunk and tail length were generally low – an order of
 619 magnitude lower than the median of 0.1% previously estimated for traits with linear
 620 measurements more generally (Houle 1992). Despite the functional relevance of trunk and tail
 621 for dispersal (McHenry and Patek 2004), our results predict that these morphological traits can
 622 only change at most a fraction of a percent per generation, even under the strongest selection
 623 possible ($\beta = 1$ or -1). For example, if there was selection for faster swimming, and selection on
 624 tail length was as strong as that on fitness ($\beta = 1$), then the evolutionary change in tail length
 625 would be 0.043% per generation. In other words, it would take about 23 ($1 / 0.043$) generations
 626 to change the population's mean phenotype by just 1%. However, the relatively short generation
 627 time (male gonads are visible through a translucent test at around 2 to 3 weeks of age in the lab

(*pers. obs.*; Grave 1933) would mean that 23 generations would take about 46 to 69 weeks. If six months (24 weeks) per year were suitable for reproduction (*pers. obs.*), it would take about 2 to 3 years to change the population's mean phenotype by 1% (or about 44 to 67 years for tail size to halve in size). These estimates are of course crude but aim to contextualize the meaning of our evolvability estimates.

A few studies have shown that higher evolvability within populations is associated with greater phenotypic divergence among populations or closely related species, suggesting the possibility that evolutionary change in contemporary populations (microevolution) can predict the pattern of evolution over ~millions of generations (macroevolution) (e.g., Holstad et al. 2024; Tsuboi et al. 2024 and references therein). If evolvability predicts phenotypic divergence, then we would expect tail size to diverge among ascidian species more than trunk size, and that fluctuating or relaxed selection on larval swimming and dispersal distance could be a major cause of such divergence. However, data collated on trunk and tail size across species in the Class Ascidiacea do not seem to support this prediction – for example, colonial ascidians have larger trunks than solitary ascidians, but similar tail lengths (McHenry and Patek 2004; colonial ascidians have tails that are smaller in proportion to trunk size than solitary species, suggesting larger maternal investment per larvae).

In contrast, within the family Molgulidae, higher evolvability in tail length does seem to be associated with greater divergence in tail length – tailless larvae have independently evolved multiple times in different geographic locations (out of 580 described species, 59 have tailless larvae; Jeffery et al. 1999; Huber et al. 2000; Fodor et al. 2021). Therefore, independent lineages have repeatedly diverged along the larval morphology axis in which we measured the highest evolvability. The main hypothesis offered for why tailless larvae evolved is that tailless larvae

are selectively advantageous when there is a need to reduce dispersal (tailless larvae also often metamorphose faster), such as when faster swimming against the current is not enough to offset strong tidal currents or wave action that sweep larvae away from suitable adult habitat (Jeffery et al. 1999; Huber et al. 2000; Fodor et al. 2021). An older hypothesis is that tailless larvae persist when selection on tail size is relaxed, such as in homogeneous habitats like sand flats where adult habitat is widely available (Berrill 1950). In tunicates, the *Hox* genes that determine anterior–posterior pattern during development are not colinearly arranged on the chromosome and their expression is also dispersed between larval and adult development (Fodor et al. 2021). *Hox* genes have been lost seemingly stochastically among tunicate lineages, reducing developmental constraints on the tail while preserving trunk patterning and potentially increasing the evolvability of tail length. As a result, divergence in tail length among species could reflect the underlying structure of the **G** matrix and fluctuating selection on swimming ability and dispersal distance (Holstad et al. 2024). As Fodor et al. (2021) suggest, it will be interesting to know if *Hox* degeneration facilitated evolvability in tail length, or occurred after relaxed selection on tail length.

The lower phenotypic and genetic variation in trunk length compared to tail length might reflect how trunks and tails grow. If tails grow primarily by increasing in length, but trunks grow primarily by increasing in volume, then a proportional increase in biomass of both trunk and tail would not change their length equally. For example, if the trunk is assumed to have the shape of a 3-dimensional ellipse (McHenry and Patek 2004), then volume would be approximated as $\frac{2}{3}\pi r^2 l$, where r is the radius (width and height) and l is the length (as we measured it). If the radius grows isometrically with length, then volume would scale with the cube of length – that is, doubling the volume would only increase length by a factor of approximately 1.26. We measured

length because directly measuring the volume of larvae accurately from 3-dimensional imaging is logistically infeasible at present for 1000's of larvae. Furthermore, converting trunk length to volume with measurements of radius would require making assumptions about the shape of trunks and how trunks develop along the width, height, and length axis. Nonetheless, our results are based on measured, rather than inferred, phenotypes and estimates of evolvability are robust on these measured scales.

It is difficult to compare our estimates of evolvability of dispersal traits with prior estimates because most studies describe evolutionary potential in terms of heritability of single traits rather than in terms of evolvability. Unsurprisingly, previous estimates of heritability of dispersal-related traits range from 0 to 1, but average 0.35 in birds and insects, and ~0.60 for marine invertebrates (Saastamoinen et al. 2018). It is well known that estimates of heritability are not comparable across studies because they do not just reflect differences in genetic variance, they also reflect differences in environmental and other sources that contribute to phenotypic variance, and they reflect differences in the units in which the phenotypes are measured (Houle 1992; Hansen et al. 2011). Nonetheless, heritability is still a poor predictor of evolvability (Houle 1992; Hansen et al. 2011). Any generality in terms of the rate at which dispersal traits evolves remains elusive until estimates of additive genetic variance for dispersal traits are mean-standardized, rather than variance-standardized. Still, only a handful of studies have estimated additive genetic variance for dispersal-related traits in marine systems, likely due in part to the difficulty of obtaining such estimates (e.g., Levin et al. 1991; Hilbish et al. 1999; Toonen and Pawlik 2001; Johnson et al. 2010; Lobón et al. 2011; Gao and Munch 2013; Aguirre et al. 2014; Schlatter et al. 2022)

One generality that does appear to be emerging is that, at least across marine taxa, maternal effects and environmental variation explain more of the phenotypic variation in dispersal than additive genetic effects. For example, maternal variance was ~9 times greater than additive genetic variance for larval length in a polychaete (Levin et al. 1991), and approximately 7 times greater for size at hatching in Atlantic silverside (Gao and Munch 2013). In our analysis, maternal effect variance was greater than additive genetic variance for trunks, tails, hatching and settling. These results are consistent with the notion that mothers can control the dispersal phenotypes of offspring through egg provisioning or other means (Zakas and Rockman 2021), even without parental care or maternal cells (like the seed coat in plants (Donohue 1999)). In ascidians, embryo development is influenced by maternal factors such as mRNAs and proteins in the oocyte (Sardet et al. 2005). The breeding design we used cannot disentangle the extent to which the maternal effect variance was due to maternal genetic or maternal environmental components. If the maternal effect variance arises from maternal genetic sources, then maternal effects can evolve and also have the potential to affect the evolution of dispersal through parent-offspring conflict (Wolf and Wade 2001; Starrfelt and Kokko 2010). The degree to which parents or offspring control dispersal is an important research agenda because both likely control dispersal in benthic marine invertebrates and fishes to some degree (Wilson et al. 2005; Zakas and Rockman 2021).

To understand the evolvability of dispersal, we need to quantify the genetic (co)variation in the multivariate set of traits that underlie dispersal outcomes for a given species. A focus on the phenotypes of organisms causing dispersal rather than just dispersal distance is necessary because the distribution of dispersal distances emerges as a consequence of phenotype-by-environment interactions, and the same phenotype might increase or decrease dispersal distance

719 depending on the environment. Despite good *a-priori* reasons for focusing on larval trunk and
720 tail sizes, it is also still possible that other traits affect dispersal in ascidians, such as swimming
721 behaviors in response to light, gravity, or current speed (Dilly 1964; Kusakabe and Tsuda 2007).
722 The difficulty of identifying relevant dispersal traits and estimating their evolvability with
723 precision should not detract from the importance of doing so – it is needed to address the grander
724 challenges of predicting the rate and direction at which dispersal evolves and explaining why the
725 distances travelled by larval stages of benthic marine adults vary by at least five orders of
726 magnitude among species, which cannot simply be explained by water currents alone (Álvarez-
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